

1. Objective

This project aims to determine the skills that are most in demand in the modern labor market by data science, analytics, and AI positions using self-scraped job postings on Indeed. These skills are then clustered into course-sized groups to design a graduate curriculum based on the observation of these market demand rather than from intuition alone. The final output is a 10-course master's curriculum that is compared across clustering methods and further validated using an external industry report and LLM-based interpretation.

2. Data Collection and Preprocessing

The analysis is based on 1,015 self-scraped Indeed postings across Canada and the United States. Initial data processing involved analysis of missingness in key fields, including usable job descriptions, which was the most critical demand in the downstream analysis since skill extraction was based on posting text. The main N-gram pipeline was applied after filtering out postings with no usable description text. The primary retained fields included job title, company, location, salary, description, and posting link, and the scraped fields were heavily missing, including rating and date, since they did not serve the purpose of the curriculum-design objective.

Raw descriptions were converted into lowercase analysis text and attached to job titles to create the corpus in which skills were extracted. A duplicate check was also conducted at the level of title, company, and description to assess repeated postings. Although 179 repeated records were identified based on title, company, and description, the main pipeline was retained without aggressive deduplication of repeated postings. Some of these repeated records may reflect the same role being reposted or promoted multiple times on Indeed, which may indicate persistent hiring demand for certain skill combinations. However, repeated postings may also inflate skill frequencies, so this pattern was acknowledged as a limitation and not interpreted as direct evidence of underlying demand.

A manual seed dictionary of 55 skills was first constructed from the course template and observed posting language; it was further expanded by 60 more skills using the Gemini API to create a final vocabulary of 115 skills. The final extraction step involved alias-based N-gram matching to allow identification of skills and their frequent variants across postings. After extraction, one skill was found to have zero frequency and was removed from the downstream analysis, leaving a final skill set of 114 skills.

3. Visual Evidence and Feature Engineering

A total of six visualization figures were generated. These visualizations indicate that Communication, Python and SQL continue to be the fundamental technical pillars of the present labor market, whereas AI, Machine Learning and associated production-related capabilities are also visible (see Appendix A, Fig. 1 and Fig. 2). The comparison of the two seniority levels demonstrates that lower rank jobs focus more on the basic technical skills, and higher rank jobs focus more on leadership, communication, and general professional skills (see Appendix A, Fig. 3). The category-level distribution also suggests that employer demand is not only limited to technical depth, but also to business-facing and professional skills (see Appendix A, Fig. 6). These trends promote a curriculum combining programming, AI, BI, experimentation, cloud platforms, governance, strategy, and professional communication.

To translate these observed labor-market patterns into clustering inputs, several posting-level variables were first engineered from the scraped metadata. Salary text was parsed into an approximate annual salary measure, job titles were converted into junior and senior indicators using regex rules, and location strings were used to create remote and Canada flags. The Canada flag was included because the curriculum was designed for a University of Toronto context, while the remote flag captured an important labor-market condition associated with digitally distributed roles. These variables were not used directly as clustering inputs on their own, but they made it possible to relate extracted skills to salary, seniority, and geography.

A skill-level engineered feature table was then constructed for the second clustering analysis. For each of the 114 retained skills, twelve engineered features were created to capture demand intensity (frequency, skill share), labor-market value (average salary, salary premium), job-context positioning (junior share, senior share, remote share, Canada share, location diversity, company diversity), and technical co-occurrence structure (Python co-occurrence, SQL co-occurrence). Together, these features represented each skill as more than a raw mention count, making the resulting feature space more suitable for clustering skills into curriculum-sized groups.

4. Clustering Models and Curriculum Comparison

Hierarchical clustering was performed using a cosine distance matrix computed from skill co-occurrence across job postings. This matrix quantified how similarly skills appeared across the labor market, with smaller distances indicating skills that tended to occur in similar postings. For example, relative to Python, the closest skills in the distance matrix included Machine Learning and ETL/ELT. Based on the raw hierarchical clustering results, the dendrogram illustrated many meaningful local relationships (Fig. 7), but the initial cut was highly imbalanced for curriculum design, with one dominant cluster containing 95 of the 114 skills. As a result, manual adjustment was applied to split the oversized grouping into smaller course-sized clusters while preserving the strongest relationships suggested by the dendrogram and the distance matrix. This adjustment followed interpretability and curriculum-design logic, such that skills that were relatively close in the hierarchical structure and corresponded to recognizable labor-market themes were grouped into course-sized clusters, including skills like programming, data engineering, and business intelligence. The final hierarchical solution produced 10 clusters with sizes ranging from 7 to 16 skills. Although the adjusted solution achieved a silhouette score of 0.0033, indicating weak compactness numerically, it still yielded the most balanced and interpretable curriculum structure for course design.

K-means clustering was performed on the 12 engineered features discussed above. The raw K-means clustering suggested an optimal $k = 10$ after being evaluated by the elbow curve and silhouette trend (Fig. 8). However, the resulting cluster structure was not well balanced for curriculum design, with cluster sizes ranging from 1 to 24 skills and two singleton clusters containing only Julia and Web Analytics respectively. Therefore, manual adjustment was applied to improve interpretability and course balance while maintaining the broader relationships suggested by the raw K-means solution. After adjustment, the final K-means curriculum consisted of 8 course-sized clusters with sizes ranging from 4 to 24 skills. The t-SNE plot was used to visualize the final K-means results (Fig. 9). Although the adjusted solution achieved a silhouette score of 0.2192, indicating stronger compactness numerically than the hierarchical solution, several clusters remained uneven and less naturally interpretable as a teachable curriculum. For this reason, the adjusted K-means solution was retained as a comparison model rather than selected as the final curriculum.

Ensemble clustering was not selected because it did not produce a clearly better curriculum than the individual methods. It combined the adjusted hierarchical labels and adjusted K-means labels through a pairwise voting scheme into a consensus similarity matrix to identify areas of agreement between the two methods. For each pair of skills, the two models were checked for whether they placed that pair in the same cluster, and the final ensemble similarity was defined as the average agreement across the two methods. This similarity matrix was then converted into an ensemble distance matrix and used as the input for a final agglomerative clustering step. To check the structure of the model agreement, the ensemble consensus similarity matrix was visualized as a heatmap (Fig. 10), where darker blocks indicated skill pairs on which both clustering methods agreed more strongly. The raw ensemble solution produced several broad modules that were broadly consistent with the major curriculum themes identified by the hierarchical and K-means solutions. However, some modules remained internally mixed, while others were too narrow to form coherent courses. Therefore, manual adjustment was applied to improve interpretability and curricular balance. After adjustment, the final ensemble curriculum was reorganized into 8 course-sized themes with sizes ranging from 9 to 20 skills.

Comparison across the three model-derived curriculum models was based primarily on the number of skills assigned to each interpreted course (Fig. 11). The hierarchical model was selected as the final curriculum, organizing 114 skills into 10 courses (Appendix B). This is because the goal of this task is not only to group related skills, but to translate those groups into a coherent curriculum pathway. The DataCamp (2026) argues that effective data and AI learning should be structured, role-relevant, practice-oriented, and reinforced over time. The hierarchical solution best supports this goal because it produces a clear progression such as from foundational programming, data handling, visualization, and statistics, to machine learning and AI, and then to governance, cloud, business, and professional capabilities. In contrast, the k-means solution contains several mixed clusters that are less suitable for sequencing into a teachable curriculum, while the ensemble solution is more useful for consensus validation than for defining an ordered learning pathway.

5. LLM Interpretation of the Final Curriculum

Gemini also interpreted the final curriculum using three prompts, and it has continually perceived the curriculum as a holistic and industry-focused data science program (See Appendix C). The curriculum holds a high level of coverage of the contemporary data science lifecycle, encompassing data engineering, ML/DL, cloud services, governance, business analysis, and professional skills. Also, Gemini observed that the curriculum might be overly general in certain sections, in particular the programming foundations and statistics classes, and proposed that more focus be directed towards responsible AI, software engineering practices, and vector databases/RAG and prompt engineering. Overall, Gemini did not consider the end curriculum as a random clustering outcome, but as a plausible and practical program model that could be narrowed to focus and updated to be more current.

6. Conclusion

In sum, the project produced and compared three data-driven curriculum models using skills extracted from Indeed, and the hierarchical-clustering solution was selected as the final curriculum because it provided the clearest and most coherent course pathway while integrating technical foundations with business, governance, and professional capabilities in line with current labor-market demand.

Reference

DataCamp. (2026). The state of data & AI literacy (4th ed.). <https://media.datacamp.com/cms/the-2026-state-of-data--ai-literacy-report-44mb.pdf>

Appendix A

Figure 1. Top 10 Overall Skills

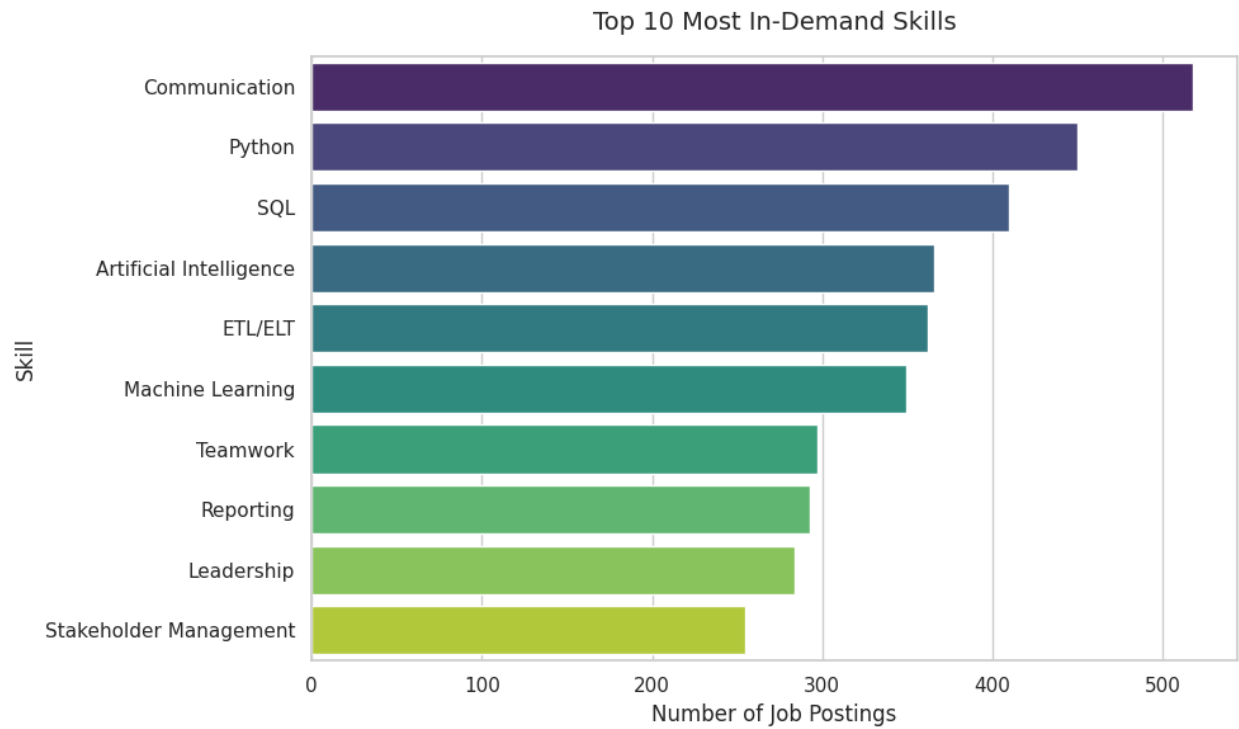


Figure 2. Skill Co-occurrence Heatmap (Top 10 Skills)

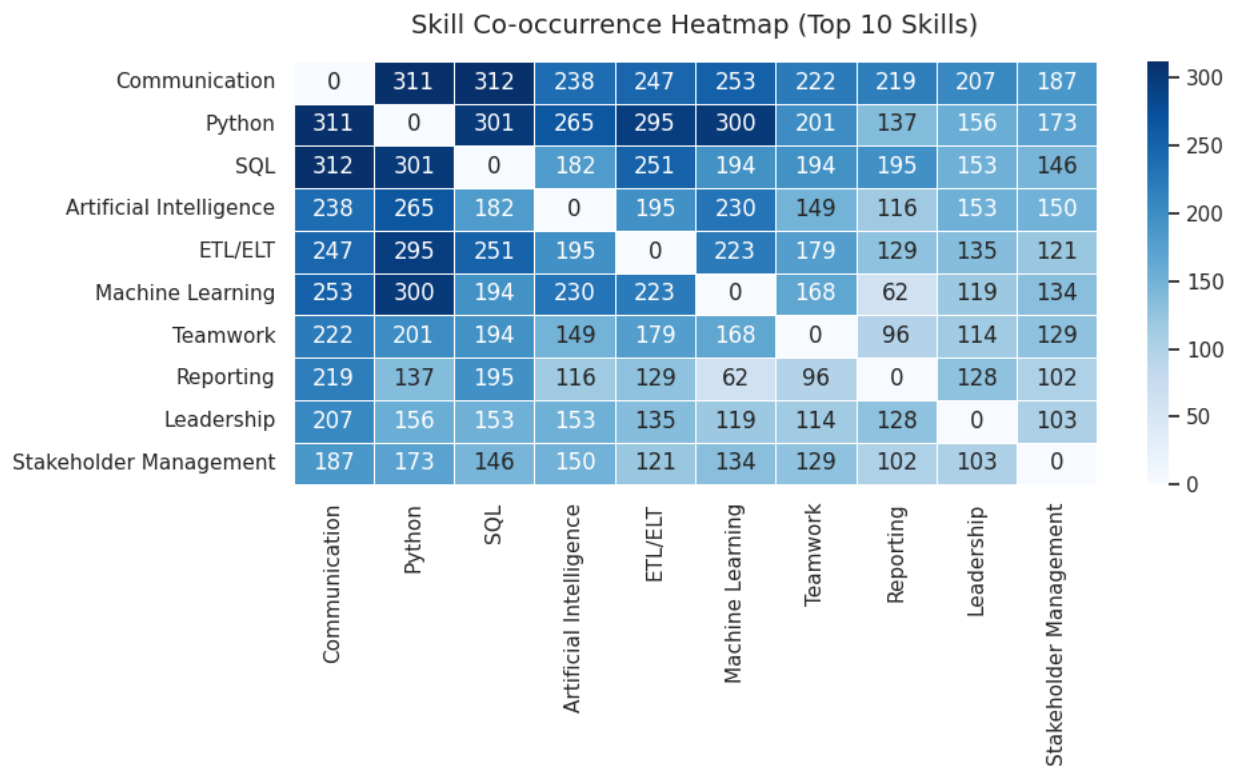


Figure 3. Percentage of Postings Requiring Top Skills: Senior vs Junior

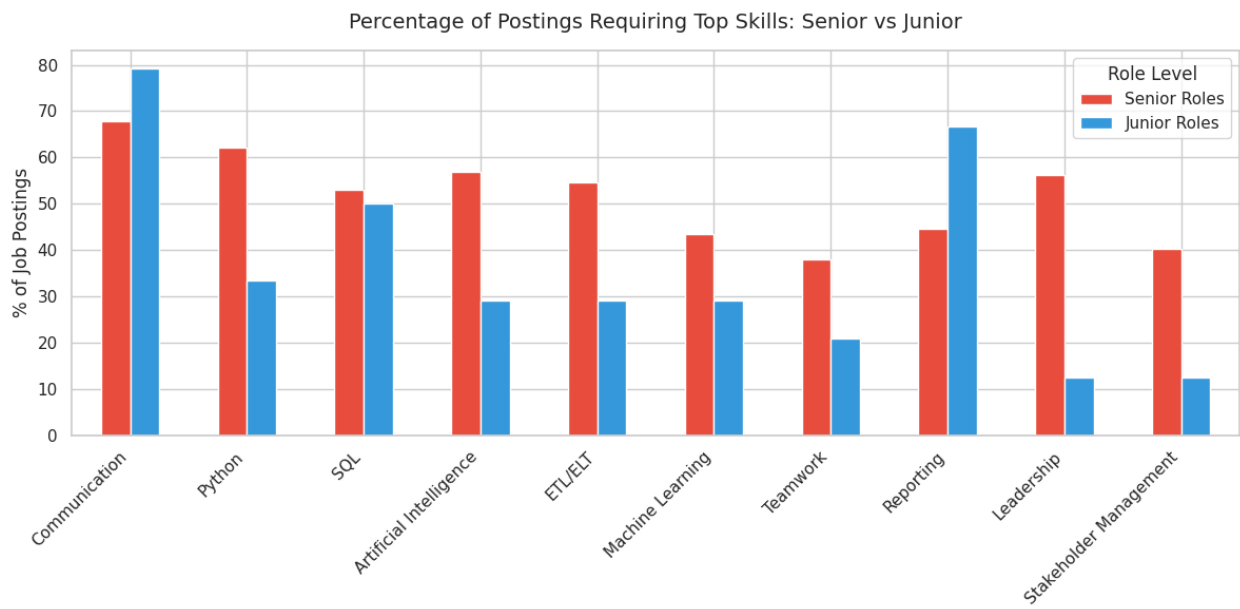


Figure 4. Top 10 Hiring Companies

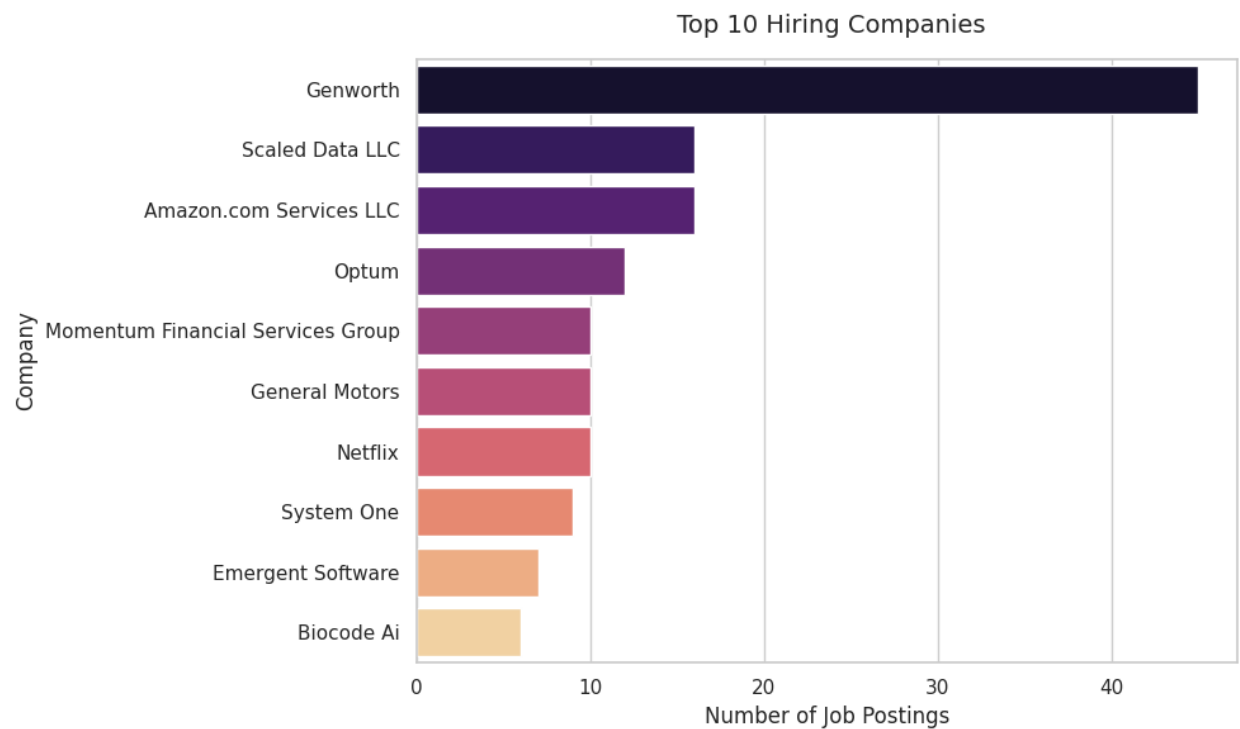


Figure 5. Top 10 Most Common Job Titles

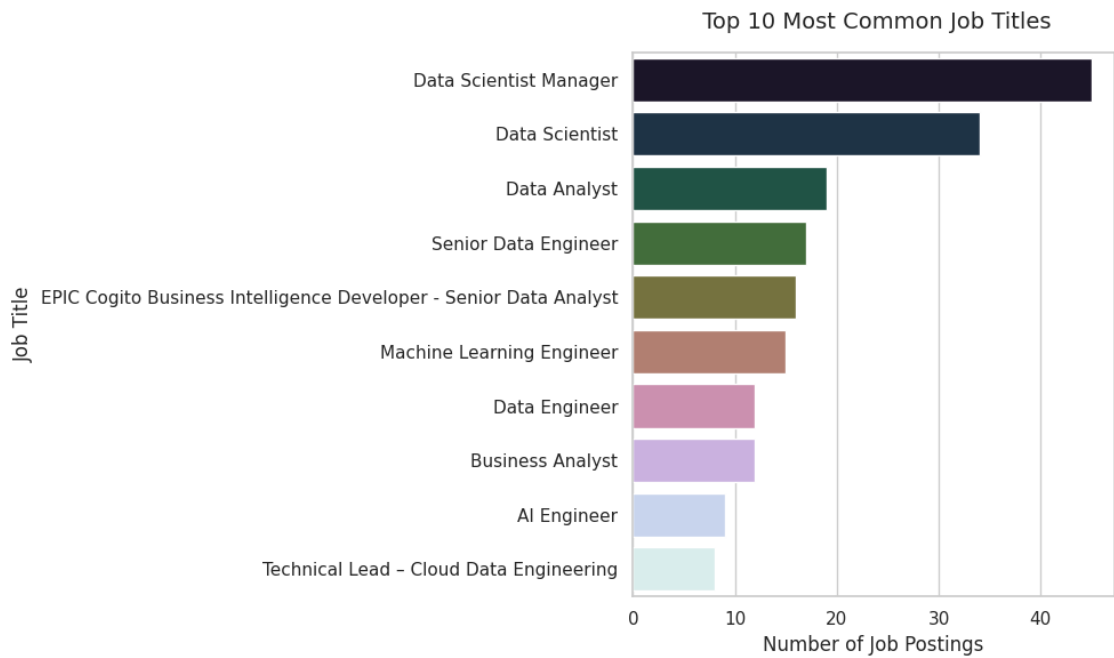


Figure 6. Distribution of Skill Categories Requested

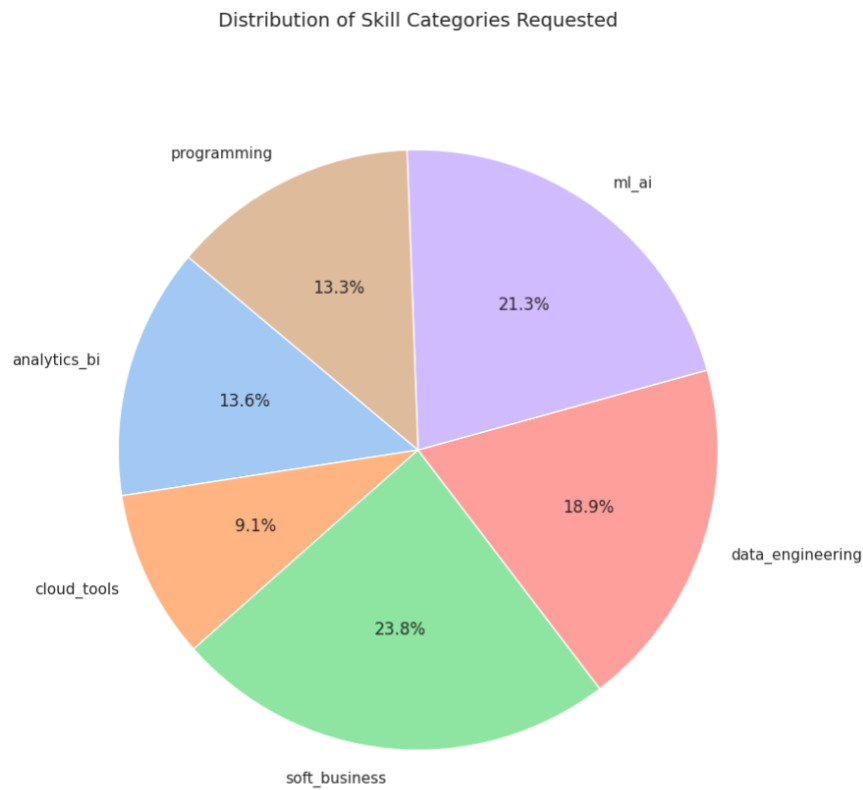


Figure 7. Hierarchical Clustering Dendrogram

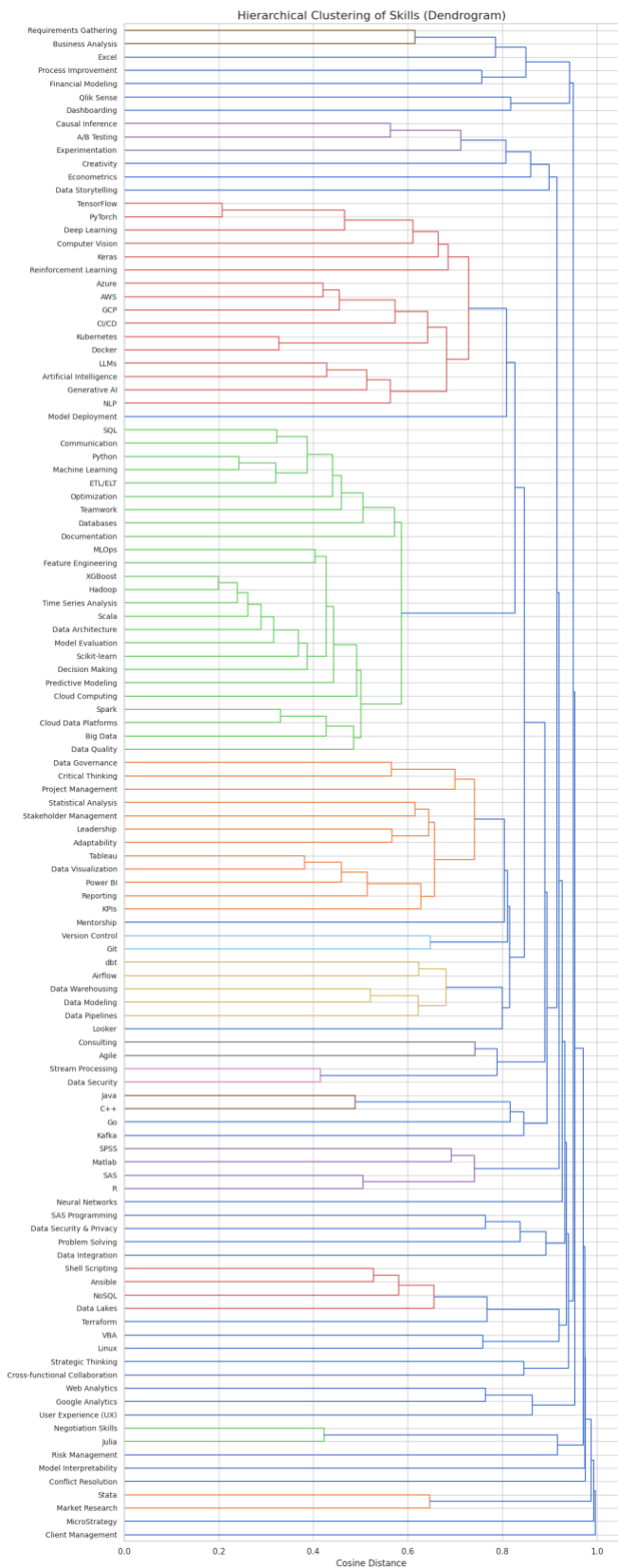


Figure 8. Elbow Curve for K-means

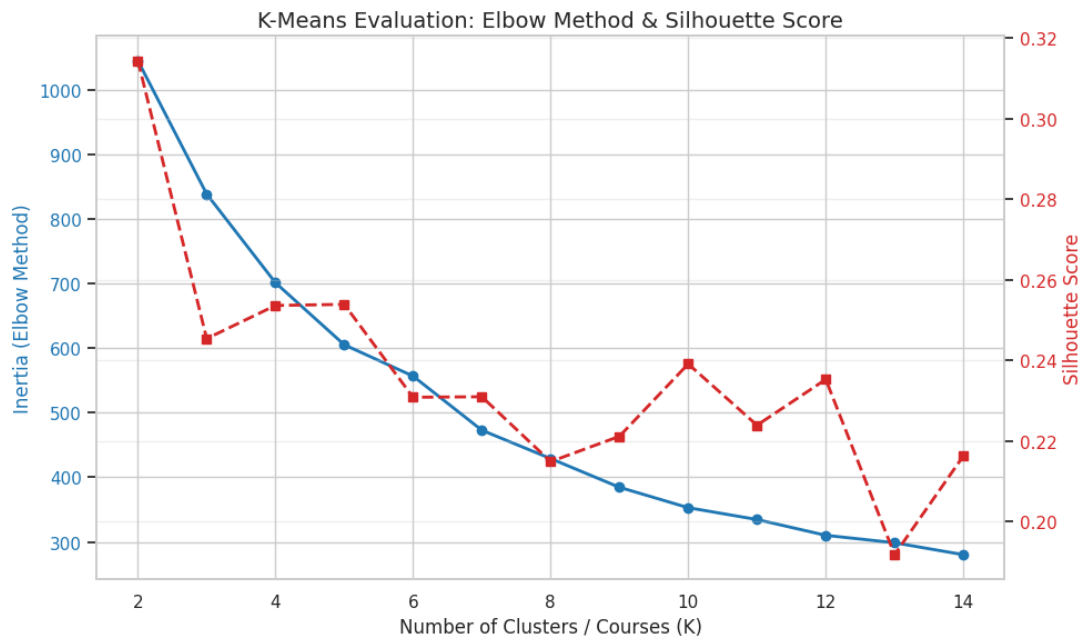


Figure 9. K-Means Clustering of Skills (t-SNE)

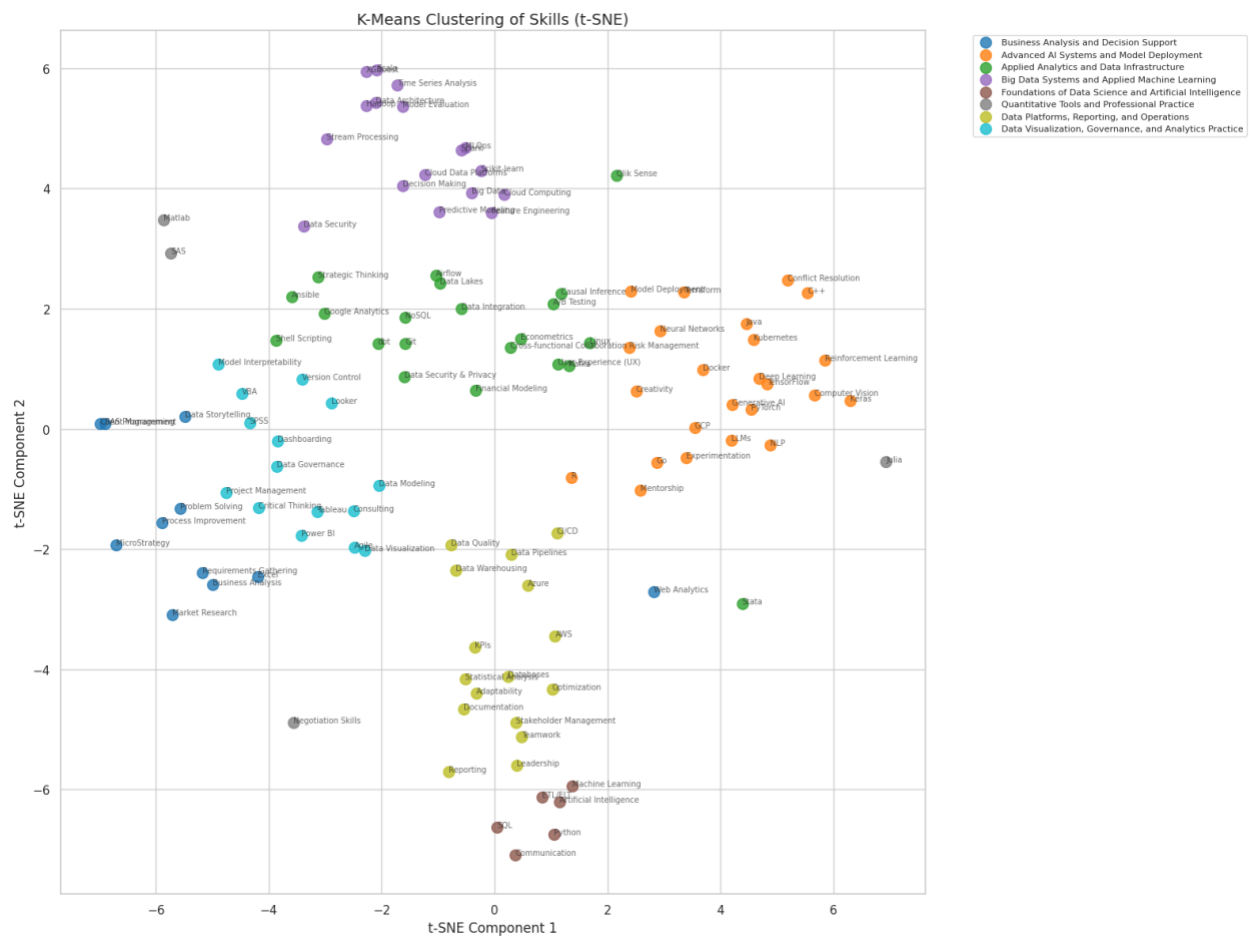


Figure 10. Ensemble Consensus Similarity Matrix

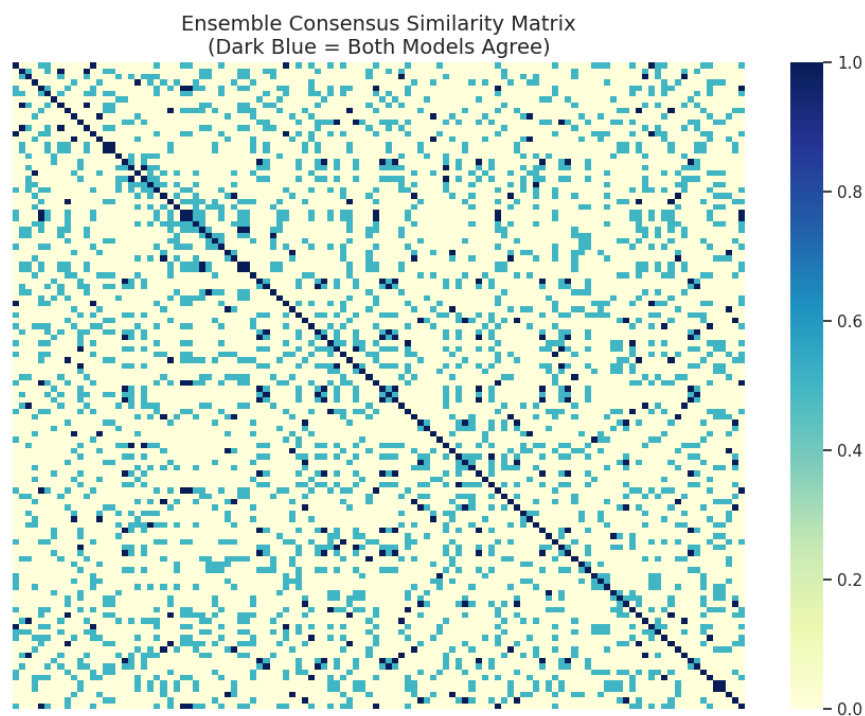
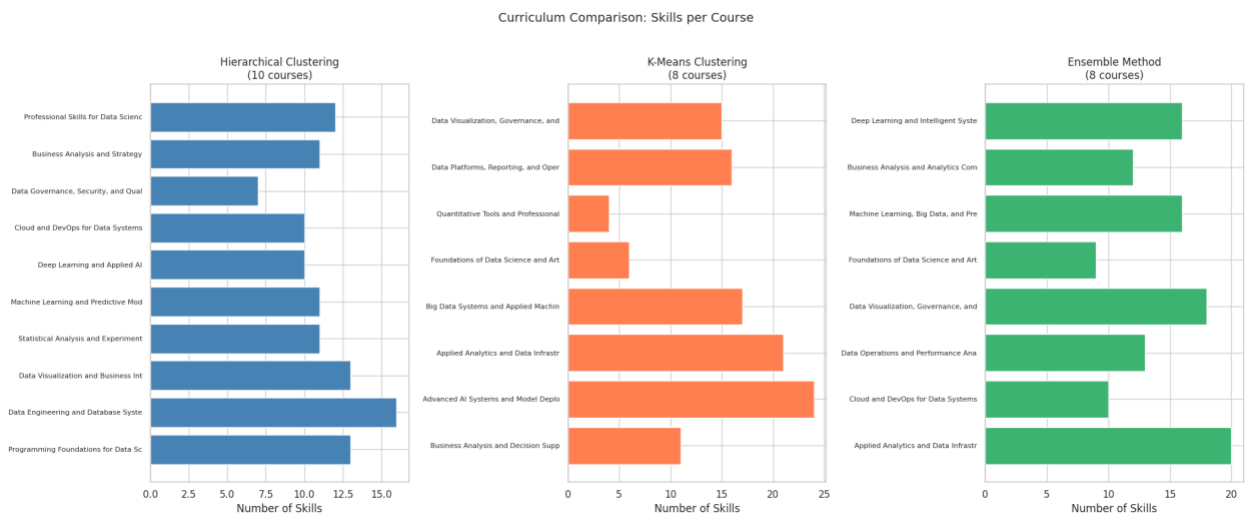


Figure 11. Curriculum Comparison Across Hierarchical, K-means, and Ensemble Methods



Appendix B

Table 1. Final 10-course curriculum

Course title	Skills covered
Programming Foundations for Data Science	Python, SQL, R, Java, Scala, C++, Go, Shell Scripting, Git, Version Control, Linux, VBA, Julia
Data Engineering and Database Systems	ETL/ELT, Data Warehousing, Data Modeling, Databases, Airflow, dbt, Spark, Kafka, Hadoop, Big Data, NoSQL, Data Lakes, Stream Processing, Data Pipelines, Data Integration, Data Architecture
Data Visualization and Business Intelligence	Tableau, Power BI, Looker, Dashboarding, Reporting, KPIs, Data Visualization, Data Storytelling, Qlik Sense, MicroStrategy, Excel, Google Analytics, Web Analytics
Statistical Analysis and Experimentation	Statistical Analysis, A/B Testing, Experimentation, Econometrics, Causal Inference, Time Series Analysis, Matlab, SPSS, SAS, SAS Programming, Stata
Machine Learning and Predictive Modeling	Machine Learning, Scikit-learn, Feature Engineering, Model Evaluation, Predictive Modeling, Optimization, MLOps, Model Deployment, XGBoost, Neural Networks, Model Interpretability
Deep Learning and Applied AI	Deep Learning, NLP, Computer Vision, LLMs, Artificial Intelligence, TensorFlow, PyTorch, Keras, Reinforcement Learning, Generative AI
Cloud and DevOps for Data Systems	AWS, Azure, GCP, Docker, Kubernetes, CI/CD, Terraform, Ansible, Cloud Computing, Cloud Data Platforms
Data Governance, Security, and Quality	Data Governance, Data Quality, Data Security, Data Security & Privacy, Requirements Gathering, Documentation, Risk Management
Business Analysis and Strategy	Business Analysis, Consulting, Agile, Strategic Thinking, Decision Making, Process Improvement, Negotiation Skills, Financial Modeling, Market Research, Client Management, User Experience (UX)
Professional Skills for Data Science and Analytics	Communication, Stakeholder Management, Leadership, Project Management, Problem Solving, Teamwork, Creativity, Critical Thinking, Adaptability, Mentorship, Cross-functional Collaboration, Conflict Resolution

Appendix C

Prompt 1 (You are a program director at University of Toronto. Based on this data science master's curriculum, write a professional 150-word description to attract students.)	Program Description As Program Director for the University of Toronto's Master of Data Science, I invite you to explore a truly comprehensive and forward-thinking curriculum. Our program meticulously builds your expertise from foundational programming (Python, SQL, R, Java) and robust data engineering (Spark, Kafka, Airflow, dbt, cloud platforms like AWS, Azure, GCP) to cutting-edge machine learning and deep learning, including LLMs and Generative AI. You will master advanced statistical analysis, A/B testing, data visualization, and business intelligence, transforming raw data into compelling narratives. Crucially, we integrate essential business analysis, strategic thinking, and professional skills—from communication and leadership to project management and ethical data governance. This holistic approach ensures you graduate not just as a technically proficient data scientist, but as a strategic leader, ready to innovate and drive impactful decisions across any industry. Response length: 968 characters
Prompt 2 (You are a Senior Data Science Hiring Manager. Audit this University of Toronto Master's curriculum against current job market trends: Please analyze: 1. Key Strengths 2. Skill Gaps 3. Overall Verdict Keep it under 250 words.)	Curriculum Validation **1. Key Strengths:** This curriculum is exceptionally robust and current. It covers the full spectrum of modern data science, from strong foundations in Python, SQL, Git, and Linux, through comprehensive data engineering (Airflow, dbt, Spark, Kafka), visualization (Tableau, Power BI), and advanced ML/DL (XGBoost, LLMs, Generative AI, MLOps). The multi-cloud exposure (AWS, Azure, GCP) and DevOps (Docker, Kubernetes) are critical. Crucially, the inclusion of Data Governance, Business Analysis, and Professional Skills (communication, leadership, project management) is outstanding, preparing

	graduates for real-world impact and senior roles. **2. Skill Gaps:** The primary concern is **Course 1's excessive list of programming languages**. Attempting to cover Python, SQL, R, Java, Scala, C++, Go, VBA, and Julia risks shallow understanding and wastes valuable time on languages not central to most modern DS roles. A sharper focus on Python, SQL, Git, and Linux is needed. Similarly, **Course 4's emphasis on outdated statistical tools like Matlab, SPSS, SAS, and Stata** should be replaced with Python/R-based methodologies to align with industry practices. **3. Overall Verdict:** Despite minor tool-specific adjustments needed in early courses, this is an **excellent, highly relevant curriculum**. It produces well-rounded data scientists with both deep technical expertise and essential business/professional acumen, making graduates exceptionally competitive for diverse roles in the current job market.
Prompt 3 (You are a data science curriculum expert. Review this Master's program curriculum and identify: 1. Important missing skills relative to current industry needs 2. Redundancies or overlapping content between courses 3. Over-emphasized areas that may not reflect current market demand Keep response under 300 words and be specific.)	Gap Analysis Here's a review of the Master's program curriculum: 1. **Important Missing Skills:** * **Software Engineering Best Practices:** Beyond basic Git, include robust testing (unit, integration), API development for data products, and advanced CI/CD for MLOps. * **Responsible AI/Ethics:** Dedicated coverage of bias detection, fairness, transparency, and the ethical implications of AI models. * **Vector Databases & RAG Architectures:** Essential for practical LLM applications, given the LLM emphasis in Course 6. * **Prompt Engineering:** A crucial practical skill for interacting with and optimizing LLMs. 2. **Redundancies/Overlapping Content:**

	* **Course 1 - Programming Foundations:** The extensive list of languages (Java, Scala, C++, Go, VBA, Julia) is excessive. Python, SQL, and R are core; others dilute focus. Java/Scala could be introduced contextually within Course 2 (Spark). * **Course 5 (Neural Networks) & Course 6 (Deep Learning):** These topics are highly overlapping. Consolidate or clearly delineate introductory neural networks from advanced deep learning applications and frameworks. * **Hadoop (Course 2):** While foundational, direct Hadoop ecosystem knowledge is less critical than cloud-native solutions or Spark, which is already covered. 3. **Over-emphasized Areas:** * **Legacy Statistical Software (Matlab, SPSS, SAS, Stata in Course 4):** Dedicating significant time to <i>all</i> these tools is inefficient. Focus on statistical <i>concepts</i> using Python/R, which are dominant in modern data science. * **Excessive BI Tools (Course 3):** Covering Tableau, Power BI, Looker, Qlik Sense, and MicroStrategy is too broad. Focus on the top 2-3 most in-demand tools for depth. * **Broad Programming Language Coverage (Course 1):** As noted, this dilutes learning on truly essential languages.
--	---

Statement of LLM Usage

LLM tools were used in a supportive role during this assignment. First, I used an LLM API to expand my initial manually constructed skill list into a broader skill dictionary for the N-gram extraction stage. Beyond that, I also used LLMs to help me understand parts of the assignment requirements, learn the basic logic of N-gram-based skill extraction, and assess the methodological soundness of my ideas for improving the web-scraping workflow. For example, I used LLMs to think through whether running multiple search combinations, merging batch CSV outputs, and enriching missing descriptions would be appropriate improvements beyond simply changing search parameters.

A Specific Example of LLM-Assisted Interpretation

One specific use of an LLM in this assignment was to help me better understand how N-gram matching works in practice for skill extraction (“can you help me understand the idea of N-Gram”), including how a manually seeded skill list could be expanded and then matched back to posting text through aliases and phrase variants.